

These processes generally have two steps—engineering change request (ECR) and engineering change order (ECO). Most PLM tools have these captured and allow the work to flow smoothly.

However, it is important to track data pertaining to failure, such as how often these failures happen or which component is the main cause. These are typically not captured or managed in a structural fashion (tools do provide options for capturing), leading to product failure. This management of end-to-end transaction is known as PLM, which is the assimilation of data from various departments like sales, manufacturing, marketing and field support to get an overall view of the product's performance. Many times, the assimilation helps in redesigning a product for extending its life in the market.

These are some issues that an SME might face. However, these are faced by established manufacturers too. In most cases, solutions to these challenges call for experts, who are good in their specialised area. In future, outsourcing will become an integral part of design, engineering and manufacturing—the level and depth of integration will depend on the specific product or business. China is a very good example of such an approach and success, and it is similar to what India achieved in the case of software.

Outsourcing options that help growth and success

To cope with the challenges, SMEs may outsource certain activities. They can review their current processes in accordance with the product engineering life cycle to help them identify the problem areas. However, most solutions effectively point to one message—any specialised job that needs special skills has to be done by the experts and with the relevant tools; compromise will lead to product failures.

For outsourcing to be successful, the process has to be understood by both the parties involved—an SME has to be prepared for a certain level of readiness and willingness to handhold the outsourcing partner, and the outsourcing partner needs to understand that it has to add

value to the product and will not be spoon fed. Some activities that can be outsourced to get maximum benefit are:

Supply chain management and component engineering. This refers to how components that make up a product are selected, procured and managed. Most big EMS vendors have dedicated departments for manning different classes of components, handling component engineering (both commodity and custom parts) and OEM procurements (like power supplies). Sometimes, established EMS vendors have special rates and delivery options with specific chip vendors. So the design team works with them directly and saves around seven to nine per cent of the cost.

Component engineering is a process using which the optimal design for a product can be selected. This process offers end-to-end data for the parts, right from the type of package and foot-print to use, expected life of the product, road map for future versions, etc. Most established EMS vendors have this as part of their business, and they help in controlling the component-related aspects impacting the product. Many component distributors have also started offering inventory management, which gives the benefits without having an in-house supply chain management team.

Printed circuit board layout. It is important to gauge the layout of a PCB from the design and manufacturing perspective. A PCB layout impacts cost, performance, manufacturing process, compliance to standards and reliability of a product. Most big OEMs are experienced and have their own design standards covering very controlled library management processes (PCB CAD tools use libraries to make the PCB layout). PCB layout was not an issue when leaded parts were used and the design was fairly straightforward. However, with the SMT, PCB layout and manufacturing processes have become interconnected. With the current-generation high-speed devices, PCB layout also contributes to EMC/EMI compliance issues. In order to get a perfect score in the assembly

line, PCB has to be laid out to suit the manufacturing process. To help clients, EMS vendors have a guideline document called DfX Guidelines, which is the intellectual property of the respective EMS vendor. The third-party PCB layout house needs to understand and implement this document. There are generic standards from IPC (Institute for Interconnecting and Packaging Electronics) like IPC 2221A and IPC-7531. However, requirements change based on the type of manufacturing machinery used and internal knowledge gained over a period, which is fed back into the respective DfX guidelines. Many times third-party design houses do not have these custom standards that impact the manufacturing process. Another issue most PCB layout houses do not understand is the current-day component characteristics (very fast rise time), and unless the PCB is laid out properly with care, the boards can fail due to fast rise time issues, as well as power integrity issues, as these high-speed devices lead to an unstable design. Most EMI/EMC problems are due to badly laid out PCBs.

Cable and harness supplies and management. Often designers think cable design does not require any special attention. Using bad cables leads to problems in the power supply section and communication ports in an otherwise well-designed product. For example, while designing a high-end camera, a team that was supposed to use miniature flex cables and connectors in the design, found out a problem with the connectors after the assembly and testing was done and the cables were ready for integration into the final product. With a little bit of research, the team found out that because of their small size, the connectors have an operational life of only 30 mating cycles, and by the time the assembly and test processes were complete, the connectors had exhausted the 30 mating cycles, leading to failure. Unless due care is given to the design and manufacture of the cable and harness, it is bound to have problems. The cable-harness fabrication needs special equipment and processes